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FILTER CIRCUIT AND PLASMA PROCESSING APPARATUS

Abstract

A filter circuit, includes: a housing made of a conductor and including an input port and an output port, formed with outer and inner conductors, wherein the housing is at ground potential with the outer conductors and includes an internal space having an area extending in first and second directions orthogonal to each other in a plane view; a partition portion made of a conductor and connected to the housing to partition the internal space; and a power feed line that is provided within the housing while insulated from the housing and includes input-side and output-side conductors, which are the inner conductors of the input and output ports, an antenna portion connected to the input-side and output-side conductors and extending in the internal space to be stacked with the partition portion, and an antenna base connecting the input-side and output-side conductors and the antenna portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-033211, filed on Mar. 5, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] This disclosure relates to a filter circuit and a plasma processing apparatus.

BACKGROUND

[0003] A plasma processing apparatus disclosed in Patent Document 1 includes a processing container in which a plasma process is performed, a stage disposed on a plate-shaped conductive base with a space disposed therebetween within the processing container to hold a substrate, a radio-frequency electrode provided in the stage, a radio-frequency power feeder for applying a radio-frequency of a certain frequency to the radio-frequency electrode, a heating element provided in the stage, a heater power feed line for electrically connecting the heating element to a heater power supply disposed at an outside of the processing container, a coil for attenuating or blocking radio-frequency noise introduced into the heater power feed line via the heating element, and filter unit including a casing for accommodating the coil.

PRIOR ART DOCUMENTS

Patent Documents

[0004] Patent Document 1: Japanese Patent Laid-Open Publication No. 2014-99585

SUMMARY

[0005] According to one embodiment of the present disclosure, a filter circuit, includes: a housing made of a conductor and including an input port and an output port, each of the input port and the output port formed with an outer conductor and an inner conductor, wherein the housing is at ground potential together with the outer conductor of the input port and the outer conductor of the output port and is configured such that an internal space, in a plane view, has an area that extends in a first direction and in a second direction orthogonal to the first direction; a partition portion made of a conductor and connected to the housing to partition the internal space; and a power feed line provided within the housing and insulated from the housing, wherein the power feed line includes: an input-side conductor, which is the inner conductor of the input port; an output-side conductor, which is the inner conductor of the output port; an antenna portion connected to the input-side conductor and the output-side conductor and extending in the internal space so as to be stacked with the partition portion; and an antenna base configured to connect the input-side conductor, the output-side conductor, and the antenna portion, and wherein the internal space has a dimension in the second direction from the antenna base and a dimension in the first direction from the antenna base larger than the dimension in the second direction from the antenna base.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the present disclosure, and together with the general description given above and the detailed description of the embodiments given below, serve to explain the principles of the present disclosure.

[0007] FIG. 1 is a schematic cross-sectional view showing an example of a configuration of a plasma processing apparatus according to an embodiment of the present disclosure.

[0008] FIG. 2 is a perspective view showing an example of a filter circuit according to the present embodiment.

[0009] FIG. 3 is a cross-sectional view showing an example of a cross-section taken along line A-A in FIG. 2.

[0010] FIG. 4 is a cross-sectional view showing an example of a cross-section taken along line B-B in FIG. 3.

[0011] FIG. 5 is a cross-sectional view showing an example of a cross-section taken along line C-C in FIG. 3.

[0012] FIG. 6 is a diagram showing an example of a simulation result of an electric field distribution viewed in a cross-section of the filter circuit according to the present embodiment.

[0013] FIG. 7 is a diagram showing an example of a simulation result of an electric field distribution in an XY plane of the filter circuit according to the present embodiment.

[0014] FIG. 8 is a graph showing an example of a frequency characteristic of the filter circuit according to the present embodiment.

[0015] FIG. 9 is a graph showing an example of a relationship between a length in an X direction of a housing and a filter frequency.

[0016] FIG. 10 is a graph showing an example of a relationship between a length in a Y direction of the housing and the filter frequency.

[0017] FIG. 11 is a graph showing an example of a relationship between the filter frequency and a characteristic parameter according to the present embodiment.

[0018] FIG. 12 is a diagram showing an example of changes in filter characteristics in contact patterns between side surfaces of the housing and a partition portion according to the present embodiment.

DETAILED DESCRIPTION

[0019] Reference will now be made in detail to various embodiments, examples of which are illustrated in the accompanying drawings. In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure.

However, it will be apparent to one of ordinary skill in the art that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, systems, and components have not been described in detail so as not to unnecessarily obscure aspects of the various embodiments.

[0020] Hereinbelow, an embodiment of a filter circuit and a plasma processing apparatus is described in detail based on the drawings. Technology disclosed herein is not limited to the following embodiment.

[0021] In a plasma processing apparatus, a power supply located outside a processing container is connected to an electrostatic chuck or a heater provided in a substrate support that supports a substrate to be processed. Since the substrate support constitutes a lower electrode for generating plasma, a radio frequency for plasma generation may affect a power feed line to the electrostatic chuck or the heater. Therefore, a radio-frequency filter including a coil and a capacitor is inserted into the power feed line. However, the radio-frequency filter including the coil and the capacitor increases structural complexity and size thereof. Therefore, when the radio-frequency filter is installed in a longitudinal direction of the power feed line, achieving a simple and compact radio-frequency filter by shortening a dimension in a width direction is expected.

[Configuration of Plasma Processing System]

[0022] Hereinafter, an example of a configuration of a plasma processing system is described. FIG. 1 is a schematic cross-sectional view showing an example of a configuration of a plasma processing apparatus according to an embodiment of the present disclosure. As shown in FIG. 1, the plasma processing system includes a capacitively coupled plasma processing apparatus 1 and a

controller **2**. The capacitively coupled plasma processing apparatus **1** includes a plasma processing chamber **10**, a gas supplier **20**, a power supply **30**, an exhaust system **40**, a direct current (DC) power supply **45**, and a filter circuit **50**. The plasma processing apparatus **1** also includes a substrate support **11** and a gas introducer. The gas introducer is configured to introduce at least one process gas into the plasma processing chamber **10**. The gas introducer includes a shower head **13**. The substrate support **11** is disposed within the plasma processing chamber **10**. The shower head **13** is disposed above the substrate support **11**. In one embodiment, the shower head **13** constitutes at least a part of a ceiling of the plasma processing chamber **10**. The plasma processing chamber **10** includes a plasma processing space **10s** defined by the shower head **13**, a sidewall **10a** of the plasma processing chamber **10**, and the substrate support **11**. The plasma processing chamber **10** includes at least one gas supply port for supplying at least one process gas to the plasma processing space **10s** and at least one gas discharge port for discharging gas from the plasma processing space **10s**. The plasma processing chamber **10** is grounded. The shower head **13** and the substrate support **11** are electrically insulated from a housing of the plasma processing chamber **10**.

[0023] The substrate support **11** includes a main body **111** and a ring assembly **112**. The main body **111** includes a central region **111a** for supporting a substrate W and an annular region **111b** for supporting the ring assembly **112**. A wafer is an example of the substrate W. The annular region **111b** of the main body **111** surrounds the central region **111a** of the main body **111** in a plane view. The substrate W is disposed on the central region **111a** of the main body **111**, and the ring assembly **112** is disposed on the annular region **111b** of the main body **111** so as to surround the substrate W on the central region **111a** of the main body **111**. Therefore, the central region **111a** is also called a substrate support surface for supporting the substrate W, and the annular region **111b** is also called a ring support surface for supporting the ring assembly **112**.

[0024] In one embodiment, the main body **111** includes a base **1110** and an electrostatic chuck **1111**. The base **1110** includes a conductive member. The conductive member of the base **1110** may serve as a lower electrode. The electrostatic chuck **1111** is disposed on the base **1110**. The electrostatic chuck **1111** includes a ceramic member **1111a** and an electrostatic electrode **1111b** disposed within the ceramic member **1111a**. The ceramic member **1111a** includes the central region **111a**. In one embodiment, the ceramic member **1111a** also includes the annular region **111b**. Another member surrounding the electrostatic chuck **1111**, such as an annular electrostatic chuck or an annular insulating member, may include the annular region **111b**. In this case, the ring assembly **112** may be disposed on the annular electrostatic chuck or the annular insulating member or may be disposed on both the electrostatic chuck **1111** and the annular insulating member. The electrostatic electrode **1111b** is connected to the DC power supply **45** through the filter circuit **50**. When a voltage from the DC power supply **45** is applied to the electrostatic electrode **1111b**, electrostatic attractive force occurs between the electrostatic chuck **1111** and the substrate W. The substrate W is attracted to the electrostatic chuck **1111** by the electrostatic attractive force that has occurred and is held by the electrostatic chuck **1111**.

[0025] In addition, at least one radio frequency (RF)/DC electrode coupled to an RF power supply **31** and/or a DC power supply **32** described later may be disposed within the ceramic member **1111a**. In this case, the at least one RF/DC electrode functions as a lower electrode. When a bias RF signal and/or a DC signal described later is supplied to the at least one RF/DC electrode, the RF/DC electrode is also called a bias electrode. In addition, the conductive member of the base **1110** and the at least one RF/DC electrode may function as a plurality of lower electrodes. Furthermore, the electrostatic electrode **1111b** may function as the lower electrode. Therefore, the substrate support **11** includes at least one lower electrode.

[0026] The substrate support **11** may include a temperature adjustment module configured to adjust at least one selected from the group of the electrostatic chuck **1111**, the ring assembly **112**, or the substrate W to a target temperature. The temperature adjustment module may include a heater, a heat transfer medium, a flow passage **1110a**, or a combination thereof. A heat transfer fluid such as

brine or gas flows through the flow passage **1110a**. In one embodiment, the flow passage **1110a** is formed within the base **1110**, and one or plural heaters are disposed within the ceramic member **1111a** of the electrostatic chuck **1111**. The substrate support **11** may include a heat transfer gas supplier configured to supply a heat transfer gas to a gap between a back surface of the substrate **W** and the central region **111a**.

[0027] The shower head **13** is configured to introduce at least one process gas from the gas supplier **20** into the plasma processing space **10s**. The shower head **13** includes at least one gas supply port **13a**, at least one gas diffusion chamber **13b**, and a plurality of gas introduction ports **13c**. The process gas supplied to the gas supply port **13a** passes through the gas diffusion chamber **13b** and is introduced into the plasma processing space **10s** from the gas introduction ports **13c**. The shower head **13** also includes at least one upper electrode. The gas introducer may include, in addition to the shower head **13**, one or plural side gas injectors (SGIs) installed in one or plural openings formed at the sidewall **10a**.

[0028] The gas supplier **20** may include at least one gas source **21** and at least one flow rate controller **22**. In one embodiment, the gas supplier **20** is configured to supply at least one process gas to the shower head **13** from each corresponding gas source **21** through each corresponding flow rate controller **22**. Each flow rate controller **22** may include, for example, a mass flow controller or a pressure-controlled flow rate controller. In addition, the gas supplier **20** may include one or more flow rate modulation devices that modulate or pulse a flow rate of the at least one process gas.

[0029] The power supply **30** includes the RF power supply **31** coupled to the plasma processing chamber **10** via at least one impedance matching circuit. The RF power supply **31** is configured to supply at least one RF signal (RF power) to at least one lower electrode and/or at least one upper electrode. Thereby, plasma is formed from the at least one process gas supplied to the plasma processing space **10s**. Therefore, the RF power supply **31** may function as at least a part of a plasma generator configured to generate plasma from one or more process gases in the plasma processing chamber **10**. Additionally, by supplying a bias RF signal to the at least one lower electrode, a bias potential may be generated on the substrate **W**, thereby drawing ion components in the formed plasma to the substrate **W**.

[0030] In one embodiment, the RF power supply **31** includes a first RF generator **31a** and a second RF generator **31b**. The first RF generator **31a** is configured to be coupled to the at least one lower electrode and/or the at least one upper electrode through the at least one impedance matching circuit and generate a source RF signal (source RF power) for plasma generation. In one embodiment, the source RF signal has a frequency in a range of 10 MHz to 300 MHz. In one embodiment, the first RF generator **31a** may be configured to generate a plurality of source RF signals having different frequencies. The generated one or more source RF signals are supplied to the at least one lower electrode and/or the at least one upper electrode.

[0031] The second RF generator **31b** is configured to be coupled to the at least one lower electrode through the at least one impedance matching circuit and generate a bias RF signal (bias RF power). A frequency of the bias RF signal may be the same as or different from the frequency of the source RF signal. In one embodiment, the bias RF signal has a frequency lower than the frequency of the source RF signal. In one embodiment, the bias RF signal has a frequency in a range of 100 kHz to 60 MHz. In one embodiment, the second RF generator **31b** may be configured to generate a plurality of bias RF signals having different frequencies. The generated one or more bias RF signals are supplied to the at least one lower electrode. In various embodiments, at least one selected from the group of the source RF signal and the bias RF signal may be pulsed.

[0032] In addition, the power supply **30** may include the DC power supply **32** coupled to the plasma processing chamber **10**. The DC power supply **32** includes a first DC generator **32a** and a second DC generator **32b**. In one embodiment, the first DC generator **32a** is configured to be connected to the at least one lower electrode and generate a first DC signal. The generated first DC signal (bias DC signal) is applied to the at least one lower electrode. In one embodiment, the

second DC generator **32b** is configured to be connected to the at least one upper electrode and generate a second DC signal. The generated second DC signal is applied to the at least one upper electrode.

[0033] In various embodiments, at least one selected from the group of the first DC signal and the second DC signal may be pulsed. In this case, a sequence of voltage pulses is applied to the at least one lower electrode and/or the at least one upper electrode. The voltage pulses may have rectangular, trapezoidal, or triangular pulse waveforms, or combinations thereof. In one embodiment, a waveform generator for generating the sequence of the voltage pulses from the DC signal is connected between the first DC generator **32a** and the at least one lower electrode.

Accordingly, the first DC generator **32a** and the waveform generator constitute a voltage pulse generator. When the second DC generator **32b** and the waveform generator constitute the voltage pulse generator, the voltage pulse generator is connected to the at least one upper electrode. The voltage pulses may have positive polarity or negative polarity. Further, the sequence of the voltage pulses may include one or plural positive voltage pulses or one or plural negative voltage pulses within one cycle. The first and second DC generators **32a** and **32b** may be provided in addition to the RF power supply **31**, or the first DC generator **32a** may be provided instead of the second RF generator **31b**.

[0034] The exhaust system **40** may be connected, for example, to a gas discharge port **10e** provided at a bottom of the plasma processing chamber **10**. The exhaust system **40** may include a pressure regulating valve and a vacuum pump. A pressure inside the plasma processing space **10s** is regulated by the pressure regulating valve. The vacuum pump may include a turbomolecular pump, a dry pump, or a combination thereof.

[0035] The filter circuit **50** removes influence of a radio-frequency power for plasma generation or bias on the DC power supply **45** when plasma is generated in the plasma processing space **10s**. The filter circuit **50** passes DC applied from the DC power supply **45** to the electrostatic electrode **1111b** and blocks the radio-frequency power flowing in a reverse direction from the electrostatic electrode **1111b**. In other words, the filter circuit **50** is provided on a power feed line (first power feed line) that feeds a power to an electrode exposed to an electromagnetic wave for generating plasma within the plasma processing chamber **10**.

[0036] The controller **2** processes a computer-executable instruction that allows the plasma processing apparatus **1** to execute various processes described in the present disclosure. The controller **2** may be configured to control each element of the plasma processing apparatus **1** to execute various processes described herein. In one embodiment, a part or all of the controller **2** may be included in the plasma processing apparatus **1**. The controller **2** may include a processor **2a1**, a storage **2a2**, and a communication interface **2a3**. The controller **2** is achieved by, for example, a computer **2a**. The processor **2a1** may be configured to perform various control operations by reading a program from the storage **2a2** and executing the read program. This program may be stored in the storage **2a2** in advance or may be acquired from a medium when necessary. The acquired program is stored in the storage **2a2** and is read from the storage **2a2** and executed by the processor **2a1**. The medium may be various storage media readable by the computer **2a** or may be a communication line connected to the communication interface **2a3**. The processor **2a1** may be a central processing unit (CPU). The storage **2a2** may include a random access memory (RAM), a read only memory (ROM), a hard disk drive (HDD), a solid state drive (SSD), or a combination thereof. The communication interface **2a3** may communicate with the plasma processing apparatus **1** via the communication line such as a local area network (LAN).

[Structure of Filter Circuit **50**]

[0037] Next, the filter circuit **50** is described in detail with reference to FIGS. **2** to **5**. FIG. **2** is a perspective view showing an example of a filter circuit according to the present embodiment. FIG. **3** is a cross-sectional view showing an example of a cross-section taken along line A-A in FIG. **2**. The following description is given by referring to a longitudinal direction of the filter circuit **50** as

an X direction along an X axis and a short direction of the filter circuit **50** as a Y direction along a Y axis. In addition, a direction along a Z axis orthogonal to the X axis and the Y axis is referred to as a Z direction. The X direction is an example of a first direction, the Y direction is an example of a second direction, and the Z direction is an example of a third direction. Cross-sections shown in FIGS. 3 to 5 illustrate that air is used as a dielectric **68** described later. As shown in FIGS. 2 and 3, the filter circuit **50** includes a housing **51**. The housing **51** is made of a conductor such as aluminum or copper. In addition, the housing **51** includes an input port **52** and an output port **55**. In the present embodiment, since the radio-frequency power is blocked, a side connected to the electrostatic electrode **1111b** is referred to as the input port **52** and a side connected to the DC power supply **45** is referred to as the output port **55**, based on a flow direction of the radio-frequency power. The input port **52** and the output port **55** may be interchangeably connected. [0038] The input port **52** and the output port **55** are formed of outer conductors **53** and **56** and inner conductors **54** and **57**, respectively. In other words, the input port **52** and the output port **55** are formed in a coaxial structure. The housing **51** is conductively connected to the outer conductors **53** and **56** and is at ground potential together with the grounded plasma processing chamber **10**, via a coaxial cable connected to the input port **52** or a frame in which the filter circuit **50** is installed. That is, the housing **51** is made of a conductor, includes the input port **52** and the output port **55** formed of the outer conductors **53** and **56** and the inner conductors **54** and **57**, respectively, and is at ground potential together with the outer conductors **53** and **56** of the input port **52** and the output port **55**. The housing **51** has, for example, a rectangular parallelepiped shape and includes a lower surface **58** and an upper surface **59** intersecting with the Z direction of the rectangular parallelepiped, side surfaces **60a** and **60b** intersecting with the X direction, and side surfaces **60c** and **60d** intersecting with the Y direction. In other words, the housing **51** is configured such that an internal space has an area in the X direction and the Y direction in a plane view (viewed in the Z direction). An edge portion of the internal space in the X and Y directions may have a chamfered, rounded shape. Also, the housing **51** is provided with the input port **52** formed on the side surface **60a** and the output port **55** formed on the side surface **60b** opposite to the side surface **60a**. The inner conductors **54** and **57** of the input port **52** and the output port **55** are respectively connected to side surfaces **63** and **64** of an upper portion **62a** of an antenna base **62** located approximately at a center of a plane in the X and Y directions in the internal space of the housing **51**. In other words, the input port **52** and the output port **55** extend in the X direction that passes through the antenna base **62**. In addition, a power feed line **61** insulated from the housing **51** is formed by the inner conductor (input-side conductor) **54** of the input port **52**, the inner conductor (output-side conductor) **57** of the output port **55**, an antenna portion **65**, and the antenna base **62**. The power feed line **61** is an example of a second power feed line and constitutes a part of a first power feed line, which is a path for feeding DC from the DC power supply **45** to the electrostatic electrode **1111b**.

[0039] The antenna base **62** connects the inner conductors **54** and **57**, which are the input-side conductor and the output-side conductor, respectively, to the antenna portion **65**. The antenna base **62** has, for example, a cylindrical shape and extends in the Z direction. The antenna portion **65** is provided at a lower surface **58** side of the inner conductors **54** and **57** in the Z direction and includes a first fin **66** and a second fin **67**. The second fin **67** includes, for example, a plurality of second fins **67a** to **67c**. The first fin **66** and the second fin **67** are made of conductors such as plate-shaped aluminum or copper and are connected to the antenna base **62** approximately at the center of the plane in the X and Y directions. The first fin **66** is, for example, thicker than each of the second fins **67a** to **67c**. The first fin **66** may also have the same thickness as each of the second fins **67a** to **67c**. That is, the antenna portion **65** is provided to be widened from the antenna base **62** in the X and Y directions. Also, the antenna portion **65** is not in contact with the four side surfaces **60a** to **60d** of the housing **51**. That is, the antenna portion **65** is a rectangle that is slightly smaller than the upper surface **59** in a plane view (when viewed in the Z direction).

[0040] A partition portion **70** made of a conductor such as aluminum or copper and connected to the housing **51** to partition the internal space of the housing **51** is provided in the internal space of the housing **51**. The partition portion **70** includes, for example, a plurality of partition portions **70a** to **70c**. The partition portion **70** is made of a conductor such as plate-shaped aluminum or copper and is connected to the side surfaces **60a** to **60d**. Also, the partition portion **70** is installed so as to have a predetermined gap (a gap **71** in FIG. **4** described later) in an annular shape with respect to the antenna base **62** of a cylindrical shape. While the partition portion **70** is in contact with the four side surfaces **60a** to **60d** of the housing **51**, it may be sufficient as long as the partition portion **70** is in contact with at least two side surfaces **60c** and **60d** intersecting with the Y direction and one side surface **60a** or **60b** intersecting with the X direction among the side surfaces **60a** to **60d** of the housing **51**. In other words, the partition portion **70** does not need to be in contact with the side surface **60a** or the side surface **60b**.

[0041] The antenna portion **65** and the partition portion **70** are provided such that the first fin **66**, the second fins **67a** to **67c**, and the partition portions **70a** to **70c** are alternately formed in the cross-section taken along line A-A. In other words, the antenna portion **65** is connected to the inner conductors **54** and **57**, which are the input-side conductor and the output-side conductor respectively, and extends into the internal space so as to be stacked with the partition portion **70**. That is, the filter circuit **50** is formed as a choke structure in which a transmission path length of spaces between the first fin **66** or the second fins **67a** to **67c** and the partition portions **70a** to **70c** or the lower surface **58** is based on a length of a $\frac{1}{4}$ wavelength of a frequency to be blocked. That is, the filter circuit **50** forms the choke structure in which spaces between the antenna portion **65** and the housing **51** or the partition portion **70**, from an end of the first fin **66** of the antenna portion **65** in the X direction, closest to the input port **52** and the output port **55**, to a center of an end **62b** of the antenna base **62** to which the second fin **67c** farthest from the input port **52** and the output port **55** is connected, are based on a length of a $\frac{1}{4}$ wavelength of an electromagnetic wave to be blocked. In the following description, the frequency to be blocked in the filter circuit **50** is also simply referred to as a filter frequency.

[0042] FIG. **4** is a cross-sectional view showing an example of a cross-section taken along line B-B in FIG. **3**. FIG. **5** is a cross-sectional view showing an example of a cross-section taken along line C-C in FIG. **3**. As shown in FIG. **4**, the partition portions **70a** to **70c** connected to the housing **51** are installed so as to have the predetermined gap **71** with respect to the antenna base **62**. On the other hand, as shown in FIG. **5**, the second fins **67a** to **67c** connected to the antenna base **62** are installed so as to have a predetermined gap **72** with respect to the side surfaces **60a** to **60d** of the housing **51**. As shown in FIGS. **2** and **3**, like the second fins **67a** to **67c**, the first fin **66** is installed so as to have a predetermined gap with respect to the side surfaces **60a** to **60d** of the housing **51**. Herein, a total propagation length (path length) of the second fins **67a** to **67c** in the X direction is desirably $\lambda/4 + \alpha$ with respect to a wavelength λ of the electromagnetic wave. Herein, α is a parameter for fine adjustment. On the other hand, a total propagation length (path length) of the second fins **67a** to **67c** in the Y direction may be sufficiently smaller than $\lambda/4$ with respect to the wavelength λ of the electromagnetic wave. For example, if the filter frequency is set to 220 MHz, dimensions in FIG. **5** may be X=55 mm and Y=25 mm. That is, outer dimensions of the housing **51** are 110 mm in the X direction and 50 mm in the Y direction.

[0043] As shown in FIG. **3**, the filter circuit **50** has a dielectric **68** between the housing **51** and the power feed line **61**. That is, the dielectric **68** is filled between the inner conductors **54** and **57** and the upper surface **59** or the side surfaces **60a** to **60d**, between the inner conductors **54** and **57** and the first fin **66**, and between the first fin **66** and the partition portion **70a**. Similarly, the dielectric **68** is also filled between the second fins **67a** to **67c** and the partition portions **70a** to **70c**, between the second fin **67c** and the lower surface **58**, and between the side surfaces **60a** to **60d** and the first fin **66** or the second fins **67a** to **67c**. In addition, when a solid material is used as the dielectric **68**, each space between the partition portions **70a** to **70c** and the first fin **66** or the second pins **67a** to

67c is configured to have the dielectric 68 having a sheet shape therebetween, so that a state in which the dielectric 68 is easily filled may be formed. Similarly, the dielectric 68 is also filled between the outer conductor 53 and the inner conductor 54 of the input port 52 and between the outer conductor 56 and the inner conductor 57 of the output port 55. The dielectric 68 may be, for example, air, polytetrafluoroethylene (PTFE), etc. That is, the dielectric 68 having a higher relative permittivity than air may be provided between the partition portion 70 and the antenna portion 65. [0044] A total propagation length (path length) of the filter circuit 50 in the X direction is a space from a tip of the first fin 66 to the end 62b of the antenna base 62 through spaces between the first fin 66, the second fins 67, and the partition portion 70. This space forms a choke structure based on a length of a $\frac{1}{4}$ wavelength of the electromagnetic wave to be blocked. The total propagation length (path length) of the filter circuit 50 in the X direction is a transmission path length W1 shown in FIG. 6. The transmission path length W1 may be expressed by Equation (2) below when the parameter α for fine adjustment described above is defined as Equation (1) below.

$$[00001] \quad \alpha = 4 \times g / 100 \quad (1) \quad W1 = g / 4 \pm \alpha \quad (2)$$

[0045] For example, if a dimension X of the second fin 67 is set to 55 mm as shown in FIG. 5, the transmission path length W1 has a one-way length of 192.5 mm or a round-trip length of 385 mm by having 3.5 round trips of 55 mm for one way. For example, if the frequency of the radio-frequency power, which is the electromagnetic wave to be blocked, is set to 220 MHz and air with a relative permittivity of 1 is used as the dielectric 68, then $2 = 1.36 \text{ m}$. Therefore, $\lambda g = \lambda / (\sqrt{1}) = 1360 \text{ mm}$, $\lambda g / 4 = 340 \text{ mm}$, and $\alpha = 54.4 \text{ mm}$. In this case, according to Equation (2), the transmission path length W1 is $340 + 54.4 \text{ mm}$. Meanwhile, when the dimension X is set to 55 mm, the transmission path length W1 is 385 mm, and this satisfies the range defined by Equation (2). When PTFE with a relative permittivity of 2.1 is used as the dielectric 68, dimensions of the first fin 66, the second fin 67, and the stacked partition portion 70 may be reduced.

[Simulation Results]

[0046] Next, simulation results of an electric field distribution of the filter circuit 50 are described with reference to FIGS. 6 and 7. FIG. 6 is a diagram showing an example of a simulation result of an electric field distribution viewed in a cross-section of the filter circuit according to the present embodiment. First, the transmission path length W1 in the cross-section of the filter circuit 50 of FIG. 6 is described. The transmission path length W1 is a path of the dielectric 68 mixed in a section 83 from a position 81 of the tip of the first fin 66 to a position 82 of the end 62b of the antenna base 62. In other words, the transmission path length W1 is a path from the position 81 of the tip of the first fin 66 to the position 82 of the end 62b, through spaces between the first fin 66 and the partition portion 70a, between the partition portion 70a and the second fin 67a, between the second fin 67a and the partition portion 70b, between the partition portion 70b and the second fin 67b, between the second fin 67b and the partition portion 70c, between the partition portion 70c and the second fin 67c, and between the second fin 67c and the lower surface 58, and via a space which is in contact with a side surface of the antenna base 62 or the side surface 60a and connects each of the spaces in the Z direction. While, in FIG. 6, a side of the input port 52 is described, a side of the output port 55 also has the same transmission path length W1. The transmission path length W1 forms a choke structure, called a $\lambda/4$ choke, based on the length of a $\frac{1}{4}$ wavelength of the electromagnetic wave to be blocked. That is, since a part of the electromagnetic wave to be blocked reciprocates along the transmission path length W1, the electromagnetic wave cancels out an electromagnetic wave (travelling wave) directly heading from the side of the input port 52 to the side of the output port 55, thereby preventing output to the side of the output port 55.

[0047] A simulation result 80 shown in FIG. 6 illustrates an electric field distribution when a radio-frequency power of 1000 W at a frequency of 220 MHz is input from the input port 52. In the simulation result 80, a space in which the dielectric 68 is present from the side of the input port 52 to the side surface 63 of the antenna base 62 shows a relatively high electric field intensity.

Additionally, in the transmission path length **W1** of a mixed path from the position **81** to the position **82**, spaces between the partition portion **70** and the first fin **66** or the second fin **67** show a relatively high electric field intensity. In a right half of FIG. **6**, the electric field distribution is obscured by an arrow of the transmission path length **W1**. The spaces between the side surface **60a** and the first fin **66** or the second fin **67** and the spaces between the partition portion **70** and the antenna base **62** show a relatively lower electric field intensity than the spaces between the partition portion **70** and the first fin **66** or the second fin **67**. Furthermore, the electric field intensity from the side surface **64** of the antenna base **62** to the side of the output port **55** is nearly zero. In other words, a reflected wave that is shifted by a phase of 180 degrees from a travelling wave is generated by the antenna portion **65** and the partition portion **70** inside the housing **51**, and the travelling wave and the reflected wave cancel each other out, such that an output of the radio-frequency power of 220 MHz from the output port **55** becomes 0 W.

[0048] FIG. **7** is a diagram showing an example of a simulation result of an electric field distribution in the XY plane of the filter circuit according to the present embodiment. The simulation result **85** shown in FIG. **7**, similar to the simulation result **80**, illustrates the electric field distribution when the radio-frequency power of 1000 W at the frequency of 220 MHz is input from the input port **52**. In the cross-section shown in FIG. **7**, a state is shown when air is used as the dielectric **68**. In the simulation result **85**, the electric field intensity becomes relatively high along the inner conductor **54** from the side of the input port **52** to the antenna base **62**. That is, the electric field intensity becomes relatively low near the side surfaces **60c** and **60d** in the XY plane, so that it may be appreciated that, even if an edge at each portion (a connection portion of the side surfaces **60a** and **60c**) in the XY plane is formed in a round shape, this does not affect the electric field distribution. Additionally, the electromagnetic wave in a space from the input port **52** through the antenna base **62** to the output port **55** may be considered to be a transverse electric (TE) wave.

[0049] FIG. **8** is a graph showing an example of a frequency characteristic of the filter circuit according to the present embodiment. A graph **90** shown in FIG. **8** represents the frequency characteristic of the filter circuit **50** by using an S-parameter S_{21} . In FIG. **8**, since a vertical axis of the graph represents S_{21} (insertion loss), an amount of attenuation increases as S_{21} approaches a negative side. Also, in FIG. **8**, a fundamental frequency (220 MHz) of the electromagnetic wave to be blocked is represented as a fundamental frequency **91** and a third harmonic (660 MHz) is represented as a third harmonic **92**. As shown in the graph **90**, the amount of attenuation of the filter circuit **50** is maximum at the fundamental frequency of 220 MHz and the insertion loss is -62 dB. The insertion loss in a range of 180 MHz to 260 MHz is below -30 dB. Additionally, the insertion loss of the filter circuit **50** is -58 dB at the third harmonic of 660 MHz. The insertion loss in a range of 640 MHz to 700 MHz is below -30 dB. In other words, the filter circuit **50** forms a band-stop filter centered on 220 MHz and 660 MHz. Thus, in the present embodiment, in a three-dimensional circuit in which the antenna portion **65** and the partition portion **70** are stacked in an internal space in which the dimension of the X direction from the antenna base **62** is larger than the dimension of the Y direction from the antenna base **62**, radio-frequency resonance of the frequency to be blocked is performed. Thereby, a simple and compact radio-frequency filter for a high-output radio-frequency power such as 1000 W in a very high frequency (VHF) band may be achieved. In addition, since the power feed line **61** is insulated from the housing **51**, an output of the DC power supply **45** may be applied to the electrostatic electrode **1111b** without ground fault. Furthermore, in the present embodiment, in a frequency range in which the insertion loss is -30 dB or less, an electromagnetic wave (single-peak waveform) with a frequency varied by using FM modulation, etc. or an electromagnetic wave (broadband waveform) with a plurality of frequencies generated as multiple tones may be blocked. In addition, in the filter circuit **50**, both the fundamental frequency (220 MHz) of the electromagnetic wave to be blocked and the third harmonic (660 MHz) may be simultaneously attenuated and blocked.

[Relationship Between Lengths of Housing in XY Directions and Filter Frequency]

[0050] Next, relationships between lengths in the XY directions of the housing and a filter frequency are described with reference to FIGS. 9 to 11. The length in the X direction and the length in the Y direction in FIGS. 9 to 11 correspond to the dimensions X and Y shown in FIG. 5, respectively. The dimension X is a length from a center of the antenna base 62 to an end of the second fin 67 near the side surface 60a, and the dimension Y is a length from a center line in the X direction passing through the input port 52 and the antenna base 62 to an end of the second fin 67 near the side surface 60c. The length in the X direction and the length in the Y direction may also take a center of the gap 72 shown in FIG. 5 (the center of the transmission path) as a reference for the ends of the second fin 67. The length in the X direction and the length in the Y direction may also be half the values of dimensions in the X direction and the Y direction of the housing 51 (lengths from the center of the antenna base 62 to the side surfaces 60a and 60c, respectively).

[0051] FIG. 9 is a graph showing an example of a relationship between the length in the X direction of the housing and the filter frequency. A graph 93 shown in FIG. 9 illustrates the relationship with the filter frequency when the length in the Y direction (dimension Y) is set to 20 mm and the length in the X direction (dimension X) is changed. As shown in the graph 93, the filter frequency tends to decrease as the length in the X direction increases, from the filter frequency of 281 MHz at the dimension X of 48 mm to the filter frequency of 219 MHz at the dimension X of 108 mm.

[0052] FIG. 10 is a graph showing an example of a relationship between the length in the Y direction of the housing and the filter frequency. A graph 94 shown in FIG. 10 illustrates the relationship with the filter frequency when the length in the X direction (dimension X) is set to 48 mm and the length in the Y direction (dimension Y) is changed. As shown in the graph 94, the filter frequency tends to decrease as the length in the Y direction increases, from the filter frequency of 363 MHz at the dimension Y of 15 mm to the filter frequency of 143 MHz at the dimension Y of 50 mm.

[0053] Now, generalizing the graphs 93 and 94 is considered. A wave number vector of a propagating electromagnetic wave in the filter circuit 50 is expressed as Equation (3) below. Based on Equation (3), a wave number k is defined as Equation (4). Therefore, an effective wavelength $\lambda_{\text{sub.eff}}$ may be expressed as Equation (5). Furthermore, based on Equation (5), a characteristic parameter β is defined as Equation (6).

[00002] $\vec{k} = \vec{k}_x \times \vec{k}_y$ (3) [0054] where $\vec{k}$ denotes a

wave number vector of a propagating electromagnetic wave, $\vec{k}_x$ denotes a wave number vector in the X direction, and $\vec{k}_y$ denotes a wave number vector in the Y direction.

[00003] $k = \sqrt{k_x^2 + k_y^2}$ (4) $\frac{1}{\lambda_{\text{eff}}} = \sqrt{\left(\frac{1}{\lambda_x}\right)^2 + \left(\frac{1}{\lambda_y}\right)^2}$ (5)

$= \sqrt{\left(\frac{1}{\lambda_x}\right)^2 + \left(\frac{1}{\lambda_y}\right)^2}$ (6)

[0055] Herein, $\lambda_{\text{sub.x}}$ is an effective wavelength in the X direction, and $\lambda_{\text{sub.y}}$ is an effective wavelength in the Y direction. Additionally, X and Y correspond to the dimensions X and Y described above, respectively. That is, the dimension X is the length from the center of the antenna base 62 to the end of the second fin 67 near the side surface 60a, and the dimension Y is the length from the center line in the X direction passing through the input port 52 and the antenna base 62 to the end of the second fin 67 near the side surface 60c.

[0056] FIG. 11 shows an example of a relationship between the filter frequency and the characteristic parameter according to the present embodiment. As shown in a graph 95 in FIG. 11, there is a linear relationship between the filter frequency and the characteristic parameter β . Thus, in the present embodiment, dimensions of the second fin 67 in the XY directions may be determined based on the graph 95 according to a desired filter frequency. That is, the internal space

of the housing **51** may have the dimension in the Y direction from the antenna base **62** and the dimension in the X direction from the antenna base **62** larger than the dimension in the Y direction from the antenna base **62**. In other words, the filter frequency in the internal space is defined by Equation (6). That is, in the present embodiment, the dimensions (external dimensions) of the housing **51** in the XY directions may be determined according to a desired filter frequency based on the graph **95**. In other words, the filter circuit **50** may reduce a dimension in a width direction (Y direction) when the filter circuit **50** is installed in a longitudinal direction (X direction) of the power feed line **61**. Namely, the filter circuit **50** may have lengths of opposing sides (lengths of the side surface **60a** or **60b** in the Y direction) at one side of the housing **51** be shorter than the length of a $\frac{1}{4}$ wavelength at the filter frequency. A difference between the dimension of the second fin **67** and the dimension of the housing **51** is set to a range that is possible to be absorbed by the fine adjustment parameter α of Equation (1) described above. In this way, in the present embodiment, the filter frequency may be universally calculated by using the characteristic parameter β .

[Contact of Side Surfaces of Housing and Partition Portion]

[0057] Next, changes in filter characteristics in contact patterns between the side surfaces **60a** to **60d** of the housing **51** and the partition portion **70** are described with reference to FIG. **12**. FIG. **12** is a diagram showing an example of the changes in the filter characteristics in the contact patterns between the side surfaces of the housing and the partition portion. In a table **96** shown in FIG. **12**, an acceptable characteristic of the filter circuit **50** is represented as a circle ($\circ$) and an unacceptable characteristic of the filter circuit **50** is represented as a cross ($\times$). Also, a vertical axis of the table **96** indicates the side surfaces **60a** and **60b** with which the partition portion **70** is in contact, and a horizontal axis of the table **96** indicates the side surfaces **60c** and **60d** with which the partition portion **70** is in contact. As shown in the table **96**, if the partition portion **70** is not in contact with any one of the surfaces **60c** and **60d**, the filter frequency (resonance frequency) is misaligned and the third harmonic is not blocked, resulting in an unacceptable characteristic. On the other hand, if the partition portion **70** is in contact with both of the side surfaces **60a** and **60b** or if the partition portion **70** is in contact with any one of the side surfaces **60a** and **60b** and a gap δ between the other side surface of the side surfaces **60a** and **60b** and the side surface **60a** or **60b** is 2 mm or less, the characteristics are in acceptable states. That is, the condition for an acceptable characteristic is that the partition portion **70** is in contact with at least three side surfaces, except for any one of the side surfaces **60a** and **60b**, and the gap δ with a non-contact side surface **60a** or **60b** is 2 mm or less. Additionally, it is desirable that the partition portion **70** is in contact with the side surfaces **60a** to **60d** (in contact with the four side surfaces). Further, the partition portion **70** may have a gap (e.g., a mesh, a punching hole, etc.) of, for example, about $\lambda g/500$ at contact portions with the side surfaces **60a** to **60d**. As an example, the gap may be formed at an edge portion in which the electric field intensity is low.

[0058] According to the present embodiment described above, the filter circuit **50** includes the housing **51**, the partition portion **70**, and the power feed line **61**. The housing **51** is made of a conductor and includes the input port **52** and the output port **55** formed of the outer conductors **53** and **56** and the inner conductors **54** and **57**, respectively. Further, the housing **51** is at ground potential together with the outer conductors **53** and **56** of the input port **52** and the output port **55** and is configured such that an internal space, in a plane view, has an area in the first direction (X direction) and in the second direction (Y direction) orthogonal to the first direction. The partition portion **70** is made of a conductor and is connected to the housing **51** to partition the internal space. The power feed line **61** is provided within the housing **51** and is insulated from the housing **51**. In addition, the power feed line **61** is configured to include the input-side conductor, which is the inner conductor **54** of the input port **52**, the output-side conductor, which is the inner conductor **57** of the output port **55**, the antenna portion **65** which is connected to the input-side conductor and the output-side conductor and extends into the internal space so as to be stacked with the partition portion **70**, and the antenna base **62** which connects the input-side conductor, the output-side

conductor, and the antenna portion **65**. The internal space has a dimension in the second direction from the antenna base **62** and a dimension in the first direction from the antenna base **62** larger than the dimension in the second direction from the antenna base **62**. As a result, a simple and compact radio-frequency filter may be realized.

[0059] In addition, according to the present embodiment, the filter frequency in the internal space is defined by Equation (6) above. In Equation (6), β denotes the characteristic parameter, X denotes the dimension in the first direction, and Y denotes the dimension in the second direction. As a result, the dimensions (outer dimensions) of the housing **51** in the XY directions according to a desired filter frequency may be determined.

[0060] In addition, according to the present embodiment, the partition portion **70** is installed so as to have the predetermined gap **71** with respect to the antenna base **62**. As a result, a transmission path may be formed between the partition portion **70** and the antenna portion **65**.

[0061] In addition, according to the present embodiment, the antenna base **62** has a cylindrical shape. As a result, the predetermined gap **71** with respect to the antenna base **62** may be easily maintained. In addition, by making the gap **71** constant, phase misalignment (polarization) in the X direction and Y direction may be suppressed, thereby stabilizing the characteristics of the filter circuit **50**.

[0062] Furthermore, according to this embodiment, the input port **52** and the output port **55** extend in the first direction that passes through the antenna base **62**. The antenna base **62** extends in the third direction (Z direction) orthogonal to the first and second directions. The antenna portion **65** is provided to be widened in the first direction and the second direction from the antenna base **62**. As a result, a simple and compact radio-frequency filter may be realized.

[0063] According to this embodiment, the partition portion **70** is in contact with the two side surfaces **60c** and **60d** intersecting with the second direction among the side surfaces **60a** to **60d** of the housing **51**. As a result, the frequency characteristic of the filter circuit **50** may be improved.

[0064] Additionally, according to the present embodiment, the internal space has a rectangular parallelepiped shape. As a result, the antenna portion **65** and the partition portion **70** may be easily formed.

[0065] Additionally, according to the present embodiment, the partition portion **70** is in contact with the four side surfaces **60a** to **60d** of the housing **51**. As a result, the frequency characteristic of the filter circuit **50** may be further improved.

[0066] Moreover, according to this embodiment, the filter circuit **50** is configured to provide the dielectric **68** with a higher relative permittivity than air between the partition portion **70** and the antenna portion **65**. As a result, a more compact radio-frequency filter may be realized.

[0067] Furthermore, according to the present embodiment, spaces between the antenna portion and the housing **51** or the partition portion **70**, from the end of the fin of the antenna portion **65** in the first direction, closest to the input port **52** and the output port **55** (first fin **66**), to the center of the end **62b** of the antenna base **62** to which the fin farthest from the input port **52** and the output port **55** (second fin **67c**) is connected, form a choke structure based on a length of a $\frac{1}{4}$ wavelength of the electromagnetic wave to be blocked. As a result, the radio-frequency power may be blocked at the frequency to be blocked and the third harmonic.

[0068] It should be noted that the embodiments disclosed herein are exemplary in all aspects and are not restrictive. The above-described embodiments may be omitted, replaced, or modified in various forms without departing from the scope and spirit of the appended claims.

[0069] While, in the above-described embodiments, the transmission path length $W1$ is set to a $\frac{1}{4}$ wavelength of the electromagnetic wave to be blocked, the transmission path length $W1$ is not limited thereto. For example, the dimensions and the number of layers of the antenna portion **65** and the partition portion **70** may be set so that the travelling wave and the reflected wave have a transmission path length that cancels each other out.

[0070] In addition, while, in the above-described embodiments, the filter circuit **50** is connected to

the electrostatic electrode **1111b** inside the electrostatic chuck **1111**, the configuration of the filter circuit **50** is not limited thereto. For example, the filter circuit **50** may be connected to a heater, which is not shown, provided inside the substrate support **11**.

[0071] In addition, while, in the above-described embodiments, the plasma processing apparatus **1** that performs processing such as etching on the substrate **W** by using the capacitively coupled plasma as a plasma source has been described as an example, the disclosed technology is not limited thereto. If it is an apparatus that performs processing on the substrate **W** by using plasma, the plasma source is not limited to the capacitively coupled plasma, and any plasma source such as inductively coupled plasma, microwave plasma, or magnetron plasma may be used.

[0072] The present disclosure may take the following configuration.

[0073] (1) A filter circuit includes: [0074] a housing made of a conductor and including an input port and an output port, each of the input port and the output port formed with an outer conductor and an inner conductor, wherein the housing is at ground potential together with the outer conductor of the input port and the outer conductor of the output port and is configured such that an internal space, in a plane view, has an area that extends in a first direction and in a second direction orthogonal to the first direction; [0075] a partition portion made of a conductor and connected to the housing to partition the internal space; and [0076] a power feed line provided within the housing and insulated from the housing, [0077] wherein the power feed line includes: [0078] an input-side conductor, which is the inner conductor of the input port; [0079] an output-side conductor, which is the inner conductor of the output port; [0080] an antenna portion connected to the input-side conductor and the output-side conductor and extending into the internal space so as to be stacked with the partition portion; and [0081] an antenna base configured to connect the input-side conductor, the output-side conductor, and the antenna portion, and [0082] wherein the internal space has a dimension in the second direction from the antenna base and a dimension in the first direction from the antenna base larger than the dimension in the second direction from the antenna base.

[0083] (2). In the filter circuit of (1), a filter frequency in the internal space is defined by Equation (A1) below:

[00004]
$$= \sqrt{\left(\frac{1}{X}\right)^2 + \left(\frac{1}{Y}\right)^2} \quad (A1) \quad [0084]$$
 where β denotes a characteristic parameter, X denotes the dimension in the first direction, and Y denotes the dimension in the second direction.

[0085] (3) In the filter circuit of (1) or (2), the partition portion is installed so as to have a predetermined gap with respect to the antenna base.

[0086] (4) In the filter circuit of any one of (1) to (3), the antenna base has a cylindrical shape.

[0087] (5). In the filter circuit of any one of (1) to (4), the input port and the output port extend in the first direction that passes through the antenna base, [0088] the antenna base extends in a third direction orthogonal to the first direction and the second direction, and [0089] the antenna portion is provided to be widened in the first direction and the second direction from the antenna base.

[0090] (6). In the filter circuit of (5), the partition portion is in contact with two side surfaces intersecting with the second direction among side surfaces of the housing.

[0091] (7). In the filter circuit of any one of (1) to (6), wherein the internal space has a rectangular parallelepiped shape.

[0092] (8). In the filter circuit of (7), the partition portion is in contact with four side surfaces of the housing.

[0093] (9). In the filter circuit of any one of (1) to (8), a dielectric with a higher relative permittivity than air is provided between the partition portion and the antenna portion.

[0094] (10). In the filter circuit of any one of (1) to (9), spaces between the antenna portion and the housing or the partition portion, from an end of a fin of the antenna portion in the first direction, closest to the input port and the output port, to a center of an end of the antenna base to which a fin farthest from the input port and the output port is connected, form a choke structure based on a

length of a $\frac{1}{4}$ wavelength of an electromagnetic wave to be blocked.

[0095] (11). A plasma processing apparatus includes: [0096] a processing container; and [0097] a filter circuit provided in a first power feed line for feeding power to an electrode exposed to an electromagnetic wave for generating plasma within the processing container, [0098] wherein the filter circuit includes: [0099] a housing made of a conductor and including an input port and an output port, each of the input port and the output port formed with an outer conductor and an inner conductor, wherein the housing is at ground potential together with the outer conductor of the input port and the outer conductor of the output port and is configured such that an internal space, in a plane view, has an area that extends in a first direction and in a second direction orthogonal to the first direction; [0100] a partition portion made of a conductor and connected to the housing to partition the internal space; and [0101] a second power feed line provided within the housing and insulated from the housing, [0102] wherein the second power feed line includes: [0103] an input-side conductor, which is the inner conductor of the input port; [0104] an output-side conductor, which is the inner conductor of the output port; [0105] an antenna portion connected to the input-side conductor and the output-side conductor and extending into the internal space so as to be stacked with the partition portion; and [0106] an antenna base configured to connect the input-side conductor, the output-side conductor, and the antenna portion, and [0107] wherein the internal space has a dimension in the second direction from the antenna base and a dimension in the first direction from the antenna base larger than the dimension in the second direction from the antenna base.

[0108] According to the present disclosure, a simple and compact radio-frequency filter may be achieved.

[0109] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosures. Indeed, the embodiments described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the disclosures. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosures.

Claims

1. A filter circuit, comprising: a housing made of a conductor and including an input port and an output port, each of the input port and the output port formed with an outer conductor and an inner conductor, wherein the housing is at ground potential together with the outer conductor of the input port and the outer conductor of the output port and is configured such that an internal space, in a plane view, has an area that extends in a first direction and in a second direction orthogonal to the first direction; a partition portion made of a conductor and connected to the housing to partition the internal space; and a power feed line provided within the housing and insulated from the housing, wherein the power feed line includes: an input-side conductor, which is the inner conductor of the input port; an output-side conductor, which is the inner conductor of the output port; an antenna portion connected to the input-side conductor and the output-side conductor and extending in the internal space so as to be stacked with the partition portion; and an antenna base configured to connect the input-side conductor, the output-side conductor, and the antenna portion, and wherein the internal space has a dimension in the second direction from the antenna base and a dimension in the first direction from the antenna base larger than the dimension in the second direction from the antenna base.

2. The filter circuit of claim 1, wherein a filter frequency in the internal space is defined by

Equation (1) below:
$$= \sqrt{\left(\frac{1}{X}\right)^2 + \left(\frac{1}{Y}\right)^2} \quad (1) \quad \text{where } \beta \text{ denotes a characteristic parameter, } X \text{ denotes}$$

the dimension in the first direction, and Y denotes the dimension in the second direction.

3. The filter circuit of claim 2, wherein the partition portion is installed so as to have a predetermined gap with respect to the antenna base.
4. The filter circuit of claim 2, wherein the antenna base has a cylindrical shape.
5. The filter circuit of claim 2, wherein the input port and the output port extend in the first direction that passes through the antenna base, wherein the antenna base extends in a third direction orthogonal to the first direction and the second direction, and wherein the antenna portion is provided to be widened in the first direction and the second direction from the antenna base.
6. The filter circuit of claim 5, wherein the partition portion is in contact with two side surfaces intersecting with the second direction among side surfaces of the housing.
7. The filter circuit of claim 2, wherein the internal space has a rectangular parallelepiped shape.
8. The filter circuit of claim 7, wherein the partition portion is in contact with four side surfaces of the housing.
9. The filter circuit of claim 2, wherein a dielectric with a higher relative permittivity than air is provided between the partition portion and the antenna portion.
10. The filter circuit of claim 2, wherein spaces between the antenna portion and the housing or the partition portion, from an end of a fin of the antenna portion in the first direction, closest to the input port and the output port, to a center of an end of the antenna base to which a fin farthest from the input port and the output port is connected, form a choke structure based on a length of a $\frac{1}{4}$ wavelength of an electromagnetic wave to be blocked.
11. The filter circuit of claim 1, wherein the partition portion is installed so as to have a predetermined gap with respect to the antenna base.
12. The filter circuit of claim 1, wherein the antenna base has a cylindrical shape.
13. The filter circuit of claim 1, wherein the input port and the output port extend in the first direction that passes through the antenna base, wherein the antenna base extends in a third direction orthogonal to the first direction and the second direction, and wherein the antenna portion is provided to be widened in the first direction and the second direction from the antenna base.
14. The filter circuit of claim 1, wherein the internal space has a rectangular parallelepiped shape.
15. The filter circuit of claim 1, wherein a dielectric with a higher relative permittivity than air is provided between the partition portion and the antenna portion.
16. The filter circuit of claim 1, wherein spaces between the antenna portion and the housing or the partition portion, from an end of a fin of the antenna portion in the first direction, closest to the input port and the output port, to a center of an end of the antenna base to which a fin farthest from the input port and the output port is connected, form a choke structure based on a length of a $\frac{1}{4}$ wavelength of an electromagnetic wave to be blocked.
17. A plasma processing apparatus, comprising: a processing container; and a filter circuit provided in a first power feed line for feeding power to an electrode exposed to an electromagnetic wave for generating plasma within the processing container, wherein the filter circuit includes: a housing made of a conductor and including an input port and an output port, each of the input port and the output port formed with an outer conductor and an inner conductor, wherein the housing is at ground potential together with the outer conductor of the input port and the outer conductor of the output port and is configured such that an internal space, in a plane view, has an area that extends in a first direction and in a second direction orthogonal to the first direction; a partition portion made of a conductor and connected to the housing to partition the internal space; and a second power feed line provided within the housing and insulated from the housing, wherein the second power feed line includes: an input-side conductor, which is the inner conductor of the input port; an output-side conductor, which is the inner conductor of the output port; an antenna portion connected to the input-side conductor and the output-side conductor and extending into the internal space so as to be stacked with the partition portion; and an antenna base configured to connect the input-side conductor, the output-side conductor, and the antenna portion, and wherein the internal

space has a dimension in the second direction from the antenna base and a dimension in the first direction from the antenna base larger than the dimension in the second direction from the antenna base.
